

Conformal Poset Prediction for Label Ranking

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Introduction. In *supervised label ranking*, an input $X \in \mathcal{X}$ is associated with a ranking $\Pi \in \mathfrak{S}_m$ of a fixed label set $\mathcal{Y} = \{1, \dots, m\}$ (Fürnkranz and Hüllermeier, 2010). Write $a \succ_\pi b$ when π ranks a before b , and identify π with $\{(a, b) : a \succ_\pi b\}$. A ranker typically returns one permutation $\hat{\sigma}(X)$. Such a point prediction suppresses uncertainty: a model may be confident that a precedes c while having little evidence for the comparison between a and b , adjacent in $\hat{\sigma}(X)$. A natural output is therefore a strict partial order, which asserts only selected pairwise preferences and leaves the remaining pairs incomparable¹. Our goal is to predict an informative partial order $R(x)$ whose asserted comparisons are simultaneously correct, that is, $\mathbb{P}\{R(X_{n+1}) \subseteq \Pi_{n+1}\} \geq 1 - \alpha$.

Conformal prediction (Vovk et al., 2005) offers finite-sample, distribution-free coverage guarantees under exchangeability. Given any ranking nonconformity score $s(x, \pi)$, define

$$C_t^s(x) = \{\pi \in \mathfrak{S}_m : s(x, \pi) \leq t\}. \quad (1)$$

Split-conformal calibration chooses $\hat{q}_s$ such that $\mathbb{P}\{s(X_{n+1}, \Pi_{n+1}) \leq \hat{q}_s\} \geq 1 - \alpha$. The unanimous pairwise content of a nonempty ranking region is $R_t^s(x) = \bigcap_{\pi \in C_t^s(x)} \pi$, a strict partial order and the largest relation respected by every ranking in the region. By construction every ranking in $C_t^s(x)$ respects this relation, that is,

$$C_t^s(x) \subseteq \text{LE}(R_t^s(x)) = \{\pi \in \mathfrak{S}_m : R_t^s(x) \subseteq \pi\},$$

its set of linear extensions. Conformal coverage of $C_{\hat{q}_s}^s(X_{n+1})$ therefore certifies all comparisons in $R_{\hat{q}_s}^s(X_{n+1})$ simultaneously with probability at least $1 - \alpha$.

Although valid, this region-first construction calibrates more than the final output requires. Complete-ranking prediction fails whenever $\Pi_{n+1} \notin C_t^s(X_{n+1})$. Partial-order prediction, by contrast, fails only if the future ranking reverses an asserted comparison of $R_t^s(X_{n+1})$. A ranking may therefore lie outside $C_t^s(x)$ while differing only on pairs that $R_t^s(x)$ leaves incomparable. Requiring coverage of the entire ranking can consequently yield an unnecessarily large threshold and, because the consensus shrinks as the threshold grows, a less informative partial order. We instead calibrate partial-order correctness directly.

Partial abstention in label ranking (Cheng et al., 2010, 2012), probe-based predictive inference (Cauchois and Duchi, 2022), conformal inference for ranking (Fermanian et al., 2025), and conformal representations of structured outputs (Zhang et al., 2025) are important (yet not exhaustive) precedents. Accordingly, we do not claim the first distribution-free partial-order predictor. We focus on the score-to-poset transformation below, its order-theoretic and efficiency characterizations, and tractable reversal geometry.

1. Incomparability here denotes abstention, not equality.

Conformal Poset Prediction. For nonempty $A \subseteq \mathfrak{S}_m$, define its *poset hull* $H(A) = \text{LE}(\bigcap_{\pi \in A} \pi)$, the linear extensions of the comparisons unanimous in A . Thus $A \subseteq H(A)$, but the enlargement preserves all pairwise information encoded by A . To characterize this hull without enumerating $C_t^s(x)$, define the cost of the cheapest counterexample to the assertion that a precedes b :

$$\kappa_{ab}^s(x) = \min_{\rho: b \succ_{\rho} a} s(x, \rho). \quad (2)$$

Then, for nonempty $C_t^s(x)$, an edge is certified exactly when no counterexample is admitted:

$$(a, b) \in R_t^s(x) \iff \kappa_{ab}^s(x) > t. \quad (3)$$

The strict inequality matters because a reversing ranking of score exactly t belongs to the closed sublevel set in (1). For an observed ranking π , let

$$s^{\text{pos}}(x, \pi) = \max_{(a, b): b \succ_{\pi} a} \kappa_{ab}^s(x). \quad (4)$$

This is the cost of the strongest comparison π contradicts, and it satisfies

$$s^{\text{pos}}(x, \pi) \leq t \iff R_t^s(x) \subseteq \pi \iff \pi \in H(C_t^s(x)). \quad (5)$$

By (5), conformal coverage of s^{pos} is exactly partial-order correctness. At fixed t , its sublevel set is $H(C_t^s(x))$, whose unanimous relation remains $R_t^s(x)$. Thus, a post-hoc hull applied after calibrating $\hat{q}_s$ changes only the represented ranking set and certifies no new comparisons; the gain comes from recalibration.

Moreover, $s^{\text{pos}}(x, \pi) \leq s(x, \pi)$ pointwise because π is feasible in each reversal problem associated with an edge it contradicts. It is the greatest pointwise minorant of s with poset-representable sublevel sets, that is, of the form $\text{LE}(P)$ for a strict partial order P .

Calibrating s^{pos} at the same order statistic of the same sample gives a quantile $\hat{q}_{\text{pos}} \leq \hat{q}_s$, by pointwise domination. Both quantiles index the same family $R_t^s(x)$, which by (3) shrinks as t grows. Thus, for every x ,

$$R_{\hat{q}_s}^s(x) \subseteq R_{\hat{q}_{\text{pos}}}^s(x), \quad \mathbb{P}\left\{R_{\hat{q}_{\text{pos}}}^s(X_{n+1}) \subseteq \Pi_{n+1}\right\} \geq 1 - \alpha. \quad (6)$$

Target alignment offers a finite-sample, samplewise dominance guarantee: it certifies every comparison that region-first calibration does, and possibly more. Strict improvement is not claimed for every sample.

For a general score s , computing the reversal costs (2) may require difficult combinatorial optimization; the transformation is therefore not uniformly efficient for arbitrary ranking scores. For Kendall distance, however, the reversal costs have the following closed form. For $s(x, \pi) = d_K(\pi, \hat{\sigma}(x))$, where $d_K(\cdot, \cdot)$ denotes the Kendall distance, i.e., the number of label pairs ordered differently by its two arguments, let $\text{rank}(c)$ be the position of label c in $\hat{\sigma}(x)$. Then $\kappa_{ab}^s(x) = (\text{rank}(b) - \text{rank}(a))_+$, and

$$s^{\text{pos}}(x, \pi) = \max_{(a, b): b \succ_{\pi} a} (\text{rank}(b) - \text{rank}(a))_+, \quad (a, b) \in R_t^s(x) \iff \text{rank}(b) - \text{rank}(a) > t.$$

The score is the largest predicted rank gap among realized inversions, computable in $O(m)$ from position arrays. If $\hat{\sigma} = A \succ B \succ C \succ D$ and $t = 1$, then $R_1^s = \{(A, C), (A, D), (B, D)\}$: adjacent comparisons are omitted, not tied. For integer $0 \leq t < m$, the output has $\binom{m-t}{2}$ certified pairs, width $\min\{m, t+1\}$, height $\lceil m/(t+1) \rceil$, and simultaneous rank interval $[\max\{1, r-t\}, \min\{m, r+t\}]$ for an item predicted at rank r .

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References

- Maxime Cauchois and John Duchi. Query-adaptive predictive inference with partial labels. *arXiv preprint arXiv:2206.07236*, 2022. doi: 10.48550/arXiv.2206.07236.
- Weiwei Cheng, Michaël Rademaker, Bernard De Baets, and Eyke Hüllermeier. Predicting partial orders: Ranking with abstention. In *Machine Learning and Knowledge Discovery in Databases*, volume 6321 of *Lecture Notes in Computer Science*, pages 215–230. Springer, 2010. doi: 10.1007/978-3-642-15880-3_20.
- Weiwei Cheng, Eyke Hüllermeier, Willem Waegeman, and Volkmar Welker. Label ranking with partial abstention based on thresholded probabilistic models. In *Advances in Neural Information Processing Systems*, pages 2510–2518, 2012.
- Jean-Baptiste Fermanian, Pierre Humbert, and Gilles Blanchard. Transductive conformal inference for full ranking. In *Advances in Neural Information Processing Systems*, 2025.
- Johannes Fürnkranz and Eyke Hüllermeier, editors. *Preference Learning*. Springer, Berlin, Heidelberg, 2010. doi: 10.1007/978-3-642-14125-6.
- Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, New York, 2005. doi: 10.1007/b106715.
- Botong Zhang, Shuo Li, and Osbert Bastani. Conformal structured prediction. In *International Conference on Learning Representations*, 2025.